

Practical Assignment 4

<u>Sr. No</u>	<u>Practical Questions</u>	<u>Type</u>
1	Create a list named fruits containing three elements: "apple", "banana", and "cherry". Print the entire list and its data type.	Easy
2	Given the list colors = ["red", "green", "blue", "yellow"], use positive indexing to print the first element and negative indexing to print the last element.	Easy
3	Create an empty list named numbers. Use the .append() method to add the integers 10, 20, and 30 to it one by one. Print the updated list.	Easy
4	Take a list animals = ["lion", "tiger", "leopard"]. Use the .insert() method to add "cheetah" at index 1 (so it becomes the second item). Print the final list.	Easy
5	Create a list hobbies = ["reading", "gaming", "swimming", "cooking"]. Use the .remove() method to delete "swimming" from the list. Print the list to verify.	Easy
6	Given the list tasks = ["dishes", "laundry", "dusting"], use the .pop() method without any arguments to remove the last item. Print both the removed item and the remaining list.	Easy
7	Create a list of 5 favorite movies. Use the len() function to calculate how many movies are in your list and print a message like "I have X movies in my list."	Easy

8	Create a list <code>scores = [45, 78, 89, 32, 66]</code> . Write a for loop that iterates through this list and prints each score on a separate line.	Easy
9	Given a list of integers <code>ages = [12, 18, 25, 14, 30]</code> , write a program that loops through the list and uses an if-else statement to print whether each person is an "Adult" (18 or older) or a "Minor".	Moderate
10	Write a program that starts with an empty list <code>user_inputs</code> . Use a while loop to ask the user to input a number. Append each number to the list. Stop the loop and print the list when the user inputs 0.	Moderate
11	Create a list <code>days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]</code> . Use list slicing to extract and print only the weekdays (Monday through Friday) into a new list.	Moderate
12	Create a list of numbers from 1 to 10. Write a loop that iterates through the list. If a number is odd, use the continue statement to skip it; otherwise, print the even numbers.	Moderate
13	Given the list <code>prices = [100, 250, 80, 400, 150]</code> , write a program to calculate the sum of all elements in the list using a for loop (do not use the built-in <code>sum()</code> function).	Moderate
14	Write a program that takes a string input from a user. If the length of the string is greater than 5, append it to a list named <code>secure_words</code> . If not, print "Word too short" and use pass as a placeholder for a failure log.	Moderate

15	Create a list <code>items = ["pen", "paper", "eraser"]</code> . Ask the user to input an index number. Use nested if-else statements to first check if the index is valid (within the range of the list). If valid, pop the item at that index and print it; if invalid, print "Index out of bounds".	Moderate
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16. The Guest List Manager

You are managing the VIP guest entryway for a tech conference. You start with an active list of checked-in guests: `checked_in = ["Alice", "Bob", "Charlie"]`. A new guest arrives at the gate.

- Ask the user to input the guest's name.
- Using an if-else statement, check if the name is already in the `checked_in` list.
- If they are already inside, print "Access Denied: Already checked in."
- If they are not in the list, use `.append()` to add them and print "Welcome to the event, [Name]!"

17. The Dynamic Undo Queue

You are coding the backend logic for a simple text editor's **Undo (Ctrl+Z)** feature. You track the user's recent actions in a list: `actions = ["Type 'Hello'", "Bold text", "Underline text"]`.

- Write a loop that simulates the user pressing "Undo".
- In each iteration, use `.pop()` to remove the *last* action performed and print "Undoing action: [Action Name]".
- The loop should automatically stop when there are no more actions left in the list (`len(actions) == 0`). Print "Canvas is clean!" at the end.

18. The Warehouse Stock Inventory

A retail warehouse tracks its aisle items sequentially in a list: `aisle = ["Shampoo", "Soap", "Toothpaste", "Conditioner", "Soap"]`. The store manager notes that "Soap" is accidentally listed twice, and they need to urgently place a new batch of "Hand Sanitizer" right at the very front of the aisle for a promotion.

- Write a program to locate and `.remove()` the first occurrence of "Soap".
- Use `.insert()` to place "Hand Sanitizer" at the absolute beginning (index 0) of the list.
- Print the finalized inventory list and the total number of items now on the aisle.

19. The Smart Temperature Monitor

A server room temperature log records hourly readings: temps = [22, 24, 28, 31, 25, 29, 33, 26]. The system triggers an emergency protocol if a temperature crosses a threshold.

- Write a program that loops through temps.
- If a temperature reading hits **30°C or higher**, print an alert message: "Critical Heat Warning: [Temperature]°C detected!" and immediately execute a break to stop checking subsequent temperatures.
- If the loop finishes checking all temperatures without hitting 30°C, use a flag or loop structure to print "All systems normal."

20. The Student Grading Bracket Splitter

A teacher has an unorganized list of student test marks: all_marks = [55, 92, 43, 76, 81, 64, 38, 88]. She wants to split these automatically into two distinct brackets: distinction_pool (marks greater than or equal to 75) and retest_pool (marks strictly less than 50).

- Create the two empty destination lists.
- Write a loop that iterates through all_marks.
- Use conditional logic (if-elif) to evaluate each score. Append the score to the distinction_pool if they excelled, or to the retest_pool if they need a re-exam.
- Marks between 50 and 74 do not fit either extreme criteria and should be skipped using a pass or ignored.
- Print both final pool lists to show the teacher her targeted groups.